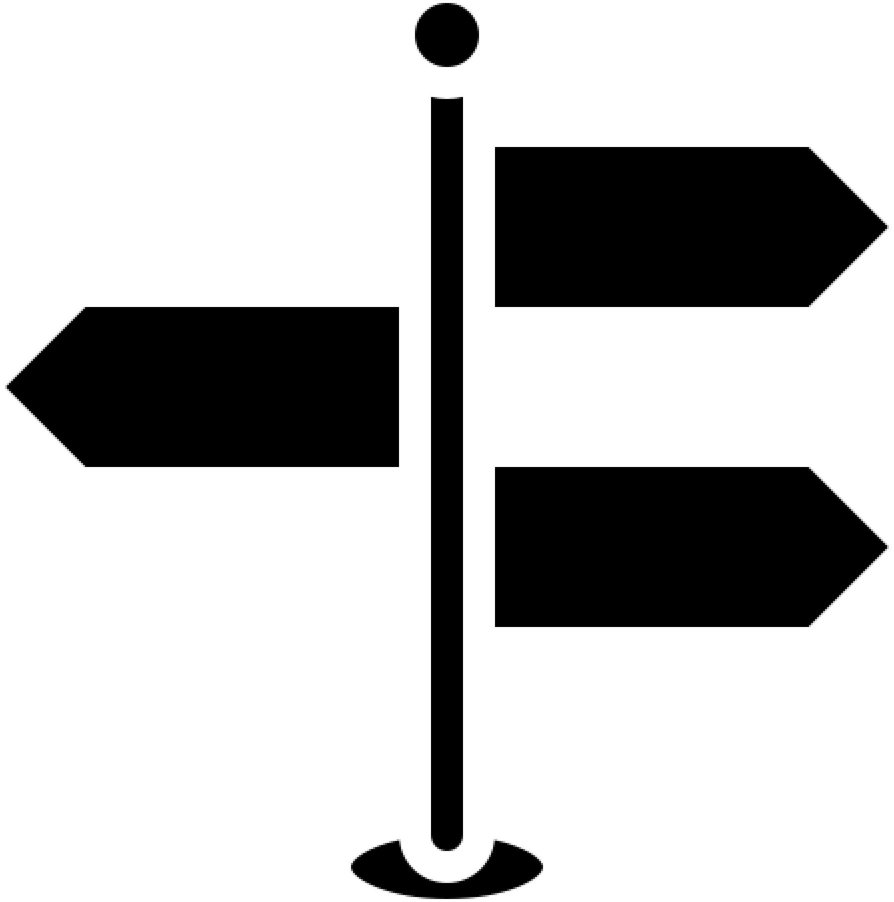
Public Cloud Services
Artificial Intelligence (AI)

Learning objectives

- Students are familiar with the basics of artificial intelligence and the services offered in the cloud.
- Students understand the implications of data protection for data storage and processing with AI services.
- Students can independently create and configure AI services.



Motivation

Gemini 2.5 Flash with the prompt “Why does humanity create AI? Visualize what comes to mind in an image.” / “Warum macht die Menscheit AI. Visualisiere was dir dazu einfällt in einem Bild.”

Agenda

- AI – Artificial Intelligence Fundamentals
- Hosting AI Solutions in the Cloud
- Cost Factors of AI Solutions
- Integration into Applications using LLMs as an Example
- Use Cases and Services

AI History

- AI has been around since the 1950s.
- The first systems were If Then, Else.
- Several AI winters between steps (no more funding).
- Various focal points, e.g., pattern recognition.
- Driving factors since 2000: availability of data and increasing computing power.

~1950	Touring Test
1980	Expert Systems
1990	Machine Learning
2000	Deep Learning & Big Data
2017	Gen AI

Ranking

AI attempts to replicate human intelligence.

- ANI = Artificial Narrow Intelligence -> Trained by humans for specific tasks.
- AGI = Artificial General Intelligence -> Independently solves tasks in new contexts.
- ASI = Artificial Super AI -> Exceeds human problem-solving abilities.

Use Cases

Optical Character Recognition oder Image Analysis

Speech to Text

Sentiment Analysis

Support Bots

ML and AI – Technologies

Frameworks

PyTorch

scikit learn

Tensorflow

Keras

Platforms

Open for Innovation KNIME

APACHE Spark™

APACHE hadoop

Hugging Face

Azure Services

- Machine Learning Studio
  - Focus on MLOps
  - Consolidation of IaaS and PaaS services for ML engineers and data scientists
  - E.g., managed Spark clusters or data stores such as OneLake
- AI Foundry & Azure OpenAI
  - Focus on LLMOps
  - Consolidation of primarily PaaS services for developers and data scientists
  - E.g., LLM inference endpoints

Machine Learning Studio

AI Foundry

Azure OpenAI

Azure Specialized AI Services

- APIs and or frameworks for use cases like speech to text or object recognition (i.e. PaaS)
  - Managed Spark Cluster
  - Managed Inference Endpoints
- Alternative hosting via VMs possible
  - VMs with GPU support
  - Free choice of framework and ML / LLM models

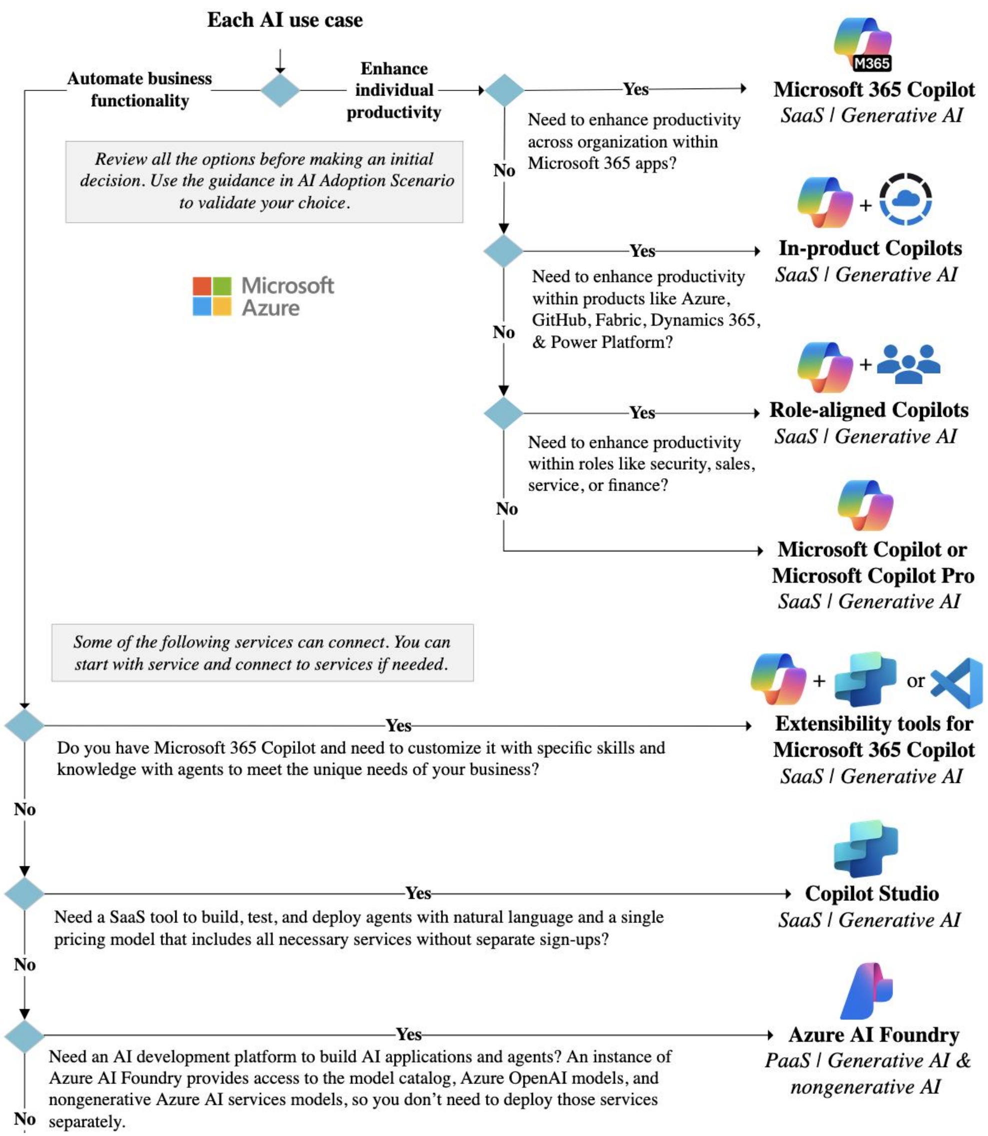
Service	Description
Azure AI Agent Service	Combine the power of generative AI models with tools that allow agents to access and interact with real-world data sources.
Azure AI Model Inference	Performs model inference for flagship models in the Azure AI model catalog.
Azure AI Search	Bring AI-powered cloud search to your mobile and web apps.
Azure OpenAI	Perform a wide variety of natural language tasks.
Bot Service	Create bots and connect them across channels.
Content Safety	An AI service that detects unwanted contents.
Custom Vision	Customize image recognition for your business.
Document Intelligence	Turn documents into intelligent data-driven solutions.
Face	Detect and identify people and emotions in images.
Immersive Reader	Help users read and comprehend text.
Language	Build apps with industry-leading natural language understanding capabilities.
Speech	Speech to text, text to speech, translation, and speaker recognition.
Translator	Use AI-powered translation technology to translate more than 100 in-use, at-risk, and endangered languages and dialects.
Video Indexer	Extract actionable insights from your videos.
Vision	Analyze content in images and videos.

Hyperscaler AI Decision Tree

Microsoft Decision Tree: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/scenarios/ai/strategy

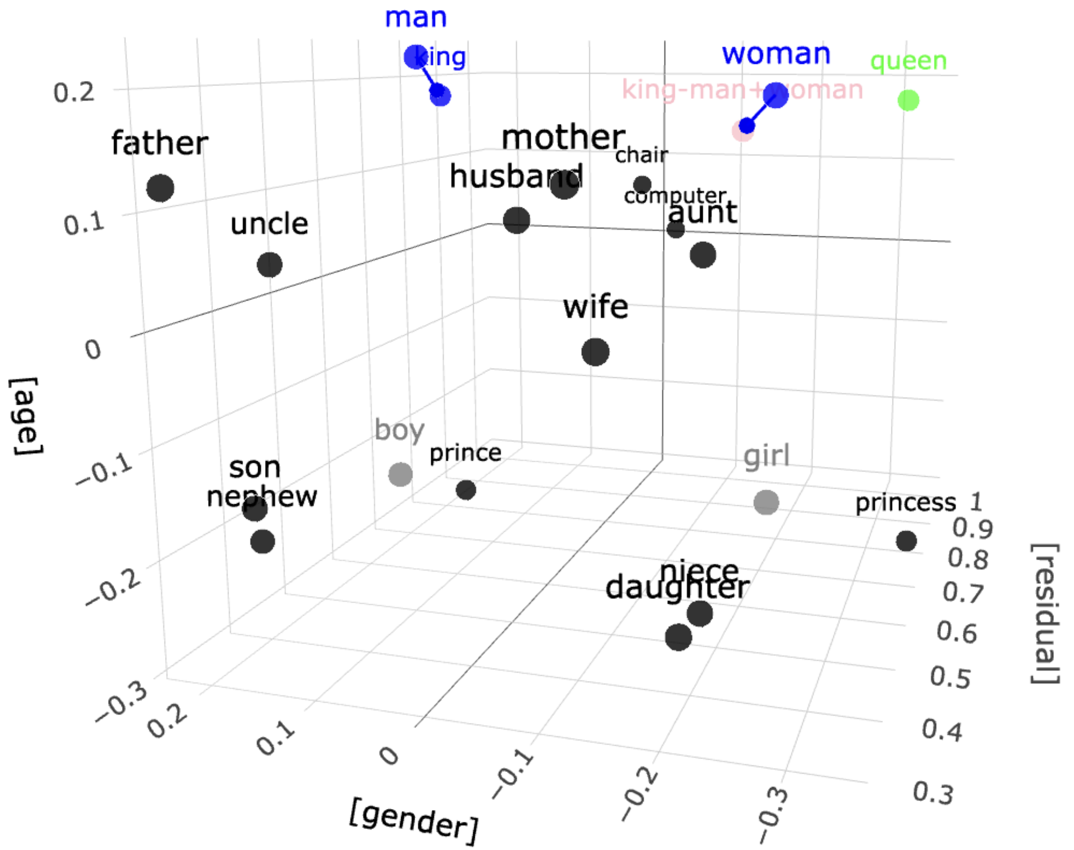
– Each hyperscaler/provider has decision trees for their AI services

– Approach: SaaS -> PaaS, depending on the use case



Large Language Models

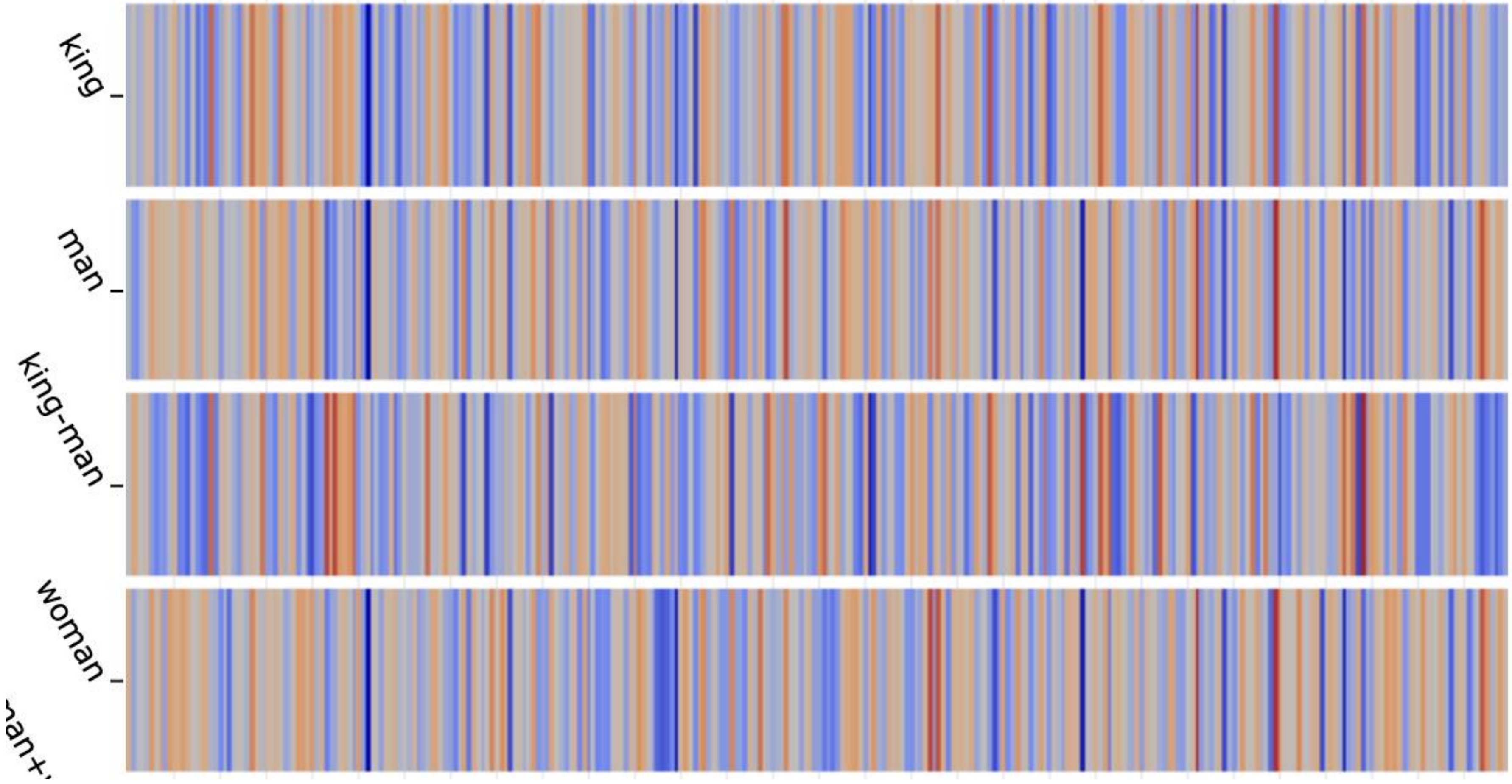
- Primary use case: Generating responses to text inputs
- LLMs are deep nural networks trained on huge amounts of data, predominantly text
- Machines can only handle numbers
- Text to token and token to text translation necessary
- Key papers:
- Efficient Estimation of Word Representations in Vector Space (Google Paper)
- Attention is all you need (Google Paper)



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/3

Large Language Models

[0,1,1,0,1,1,1,1,0,1,1,0,0,1,1,1,0,1,0,0,0,1,0,0,1,1,1,0,1,0,1,1,0,0,1,0,0,1,1,1,1,1,1,1,0,1,0,1,1,0,0,1]



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/3

Large Language Models

king − man + woman = queen

▼ Vector analogy arithmetic

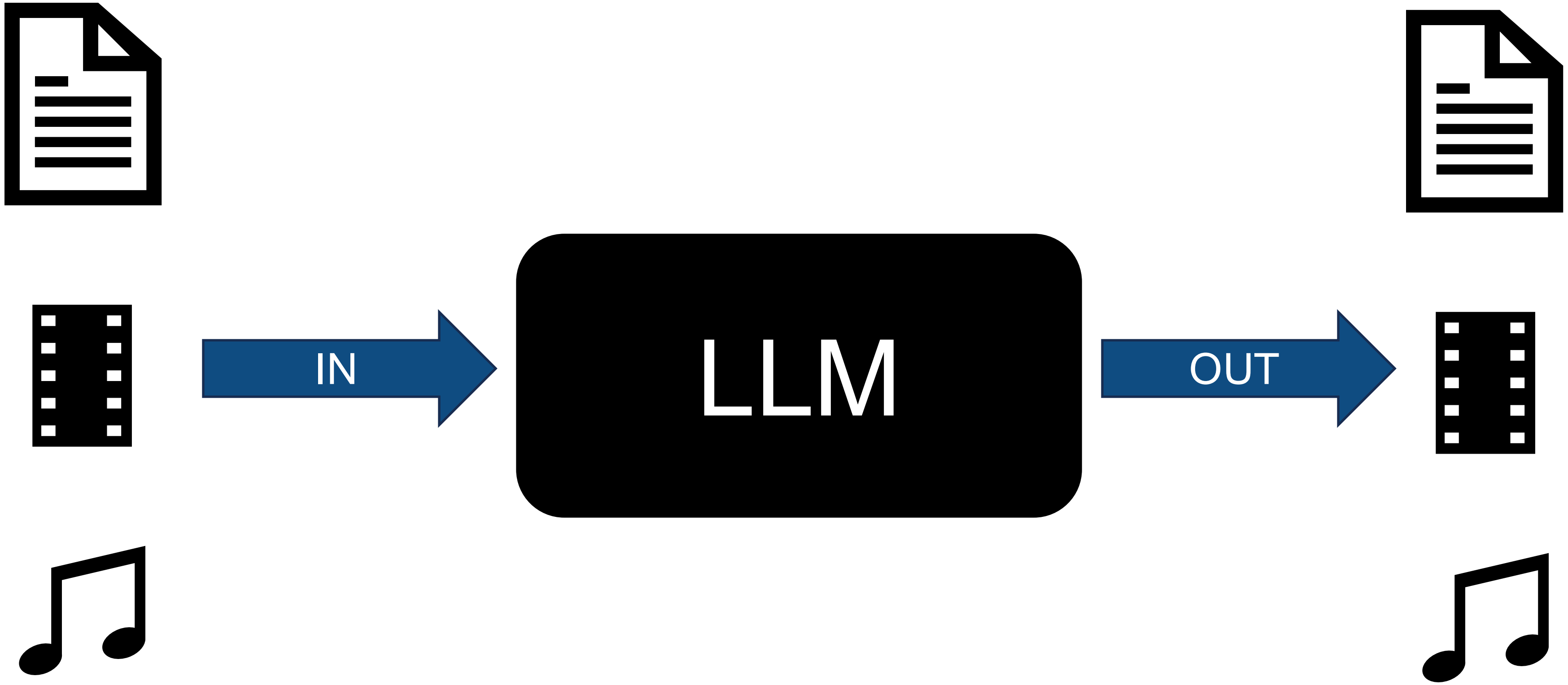
English notation						Result	
man	is to	king	as	woman	is to	queen	Submit
Vector notation						**Result**	
king	−	man	+	woman	=	queen	Submit

https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/3

Context Awareness for LLMs

Das Kleid der Frau ist Grün

Multi-Modality in LLMs



LLM Wording

Training

Calculation of model weights based on a data set.

Prompt

User query similar to the Google search bar, with or without instructions.

Context Window

A natural limit of a model in terms of the amount of tokens it can process.

Tokens

Representation of user inputs (e.g., prompts and messages) as a sequence of characters. Tokens are not the same as words.

Inference

Execution of a model with a prompt. Can be done on GPUs or CPUs.

RAG

Retrieval Augmented Generation: Enriching the model context with additional information from, for example, documents.

LLM Models

Open-Source Models	Open-Weight Models	Commercial Models
Public properties: - Model Card - Training Data - Training Code - Model Weights Examples: - Apertus - Granite	Public properties: - Model Card - Model Weights Examples: - Llama - GPT OSS - Gemma - Weitere	Public properties: - Model Card Examples: - o3, o4 etc. - Gemini Flash / Pro - Claude Sonnet / Opus - Others

LLM Services

Model Registry

Training & Optimization

Build

Monitor & Tracing

RAG

AI-Agents

Model Hosting & Inference

Guardrails

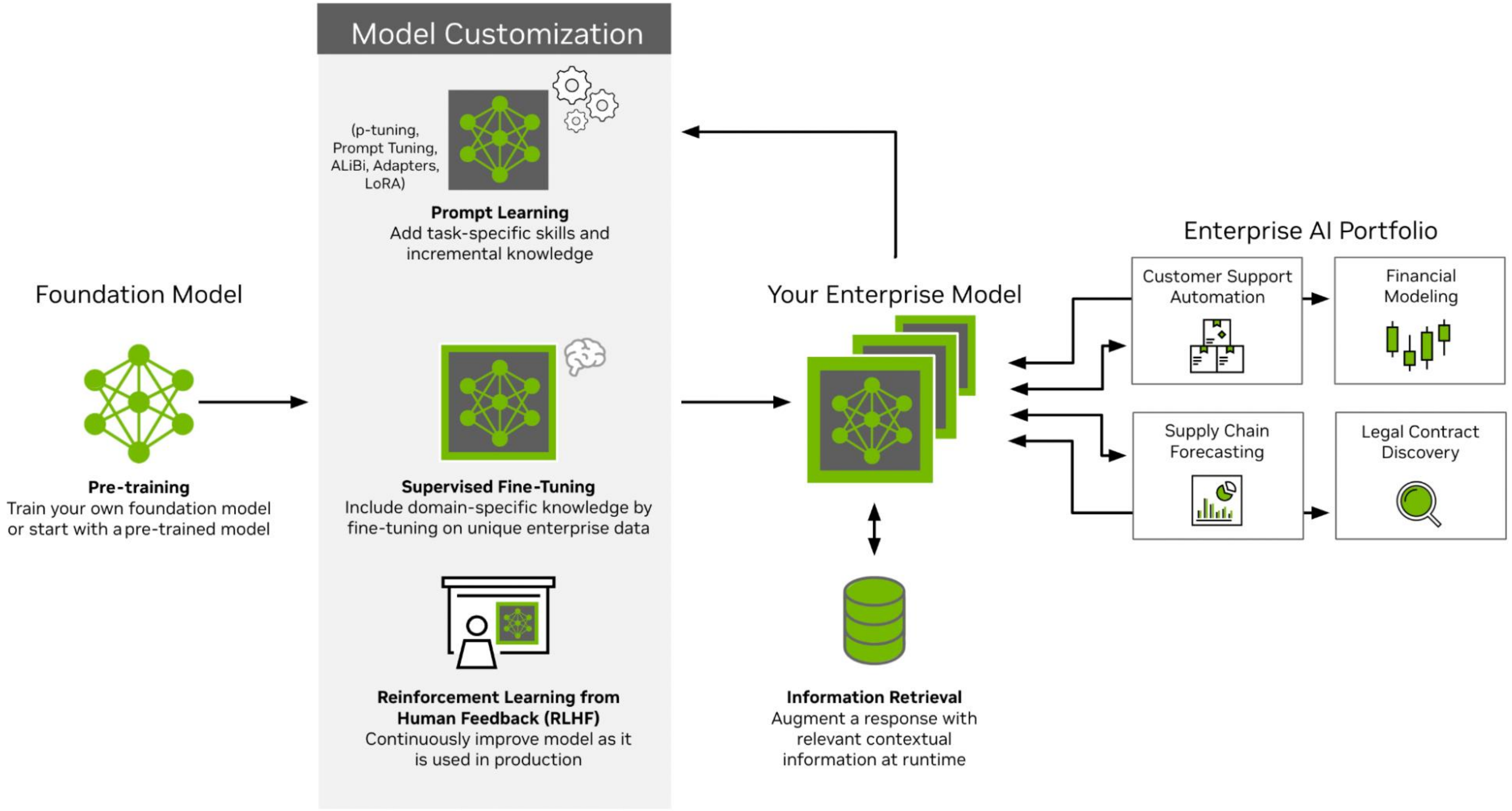
LLM Gateways

Run

AI platforms offer these services centrally and in one place. Examples include Google Vertex AI and Azure AI Foundry.

LLMOps

- Similar to MLOps, but optimized for LLMs
- Foundational models are pre-trained, generic models (cost optimization, no need to train a model from scratch)
- Training the foundational model with your own data (e.g., domain-specific)



https://developer.nvidia.com/blog/mastering-llm-techniques-llmops/

Pricing and Cost Factors for ML Services

- Billing for machine learning compute at the IaaS level, i.e., VMs
- Optional: Additional costs for service usage
- Cost factors: Runtime of compute instances and required storage capacity for data sets

General purpose

Balanced CPU-to-memory ratio. Ideal for testing and development, small to medium databases, and low to medium traffic web servers.

D2-64 v3

Instance	vCPU(s)	RAM	Linux VM Price	Machine Learning Service Surcharge	Pay As You Go Total Price	1 year savings plan	3 year savings plan
D2 v3	2	8 GiB	CHF 70.733/month	CHF 0/month	CHF 70.733/month	CHF 53.76/ month ~24% savings	CHF 38.91/ month ~45% savings

https://azure.microsoft.com/en-gb/pricing/details/machine-learning/#pricing

Pricing and Cost Factors for AI Services

- Billing for PaaS-level services
- Throughput units or pay as you go

Azure AI Speech

Commitment Tiers – Azure - Standard

Category	Features	Price (per month)	Overage
Speech to Text	Standard	CHF 1,291.921 for 2,000 hours	CHF 0.646 per hour
		CHF 5,248.426 for 10,000 hours	CHF 0.525 per hour
		CHF 20,186.251 for 50,000 hours	CHF 0.404 per hour

Azure OpenAI Service

GPT-4.1 series

GPT-4.1 series is a highly advanced general-purpose model with extensive world knowledge and an enhanced ability to understand user intent, making it particularly adept at creative tasks and agentic planning. The series features a 1 million token context window and has a knowledge cutoff of June 2024.

Model	Pricing (1M Tokens)	Pricing with Batch API (1M Tokens)
GPT-4.1-2025-04-14 Global	Input: CHF 1.62 Cached Input: CHF 0.41 Output: CHF 6.46	Input: CHF 0.81 Output: CHF 3.23

https://azure.microsoft.com/en-us/pricing/details/cognitive-services/openai-service/ und https://azure.microsoft.com/en-us/pricing/details/cognitive-services/speech-services/

Example: Calculate Cost for LLM Hosting

Comparing VM based hosting with Inference Endpoints (in Azure)

- Llama 3.3 70B Instruct model (see Huggingface)
- Compare cost per 1k token per hosting option

Example: Calculate Cost for LLM Hosting – Azure AI Foundry

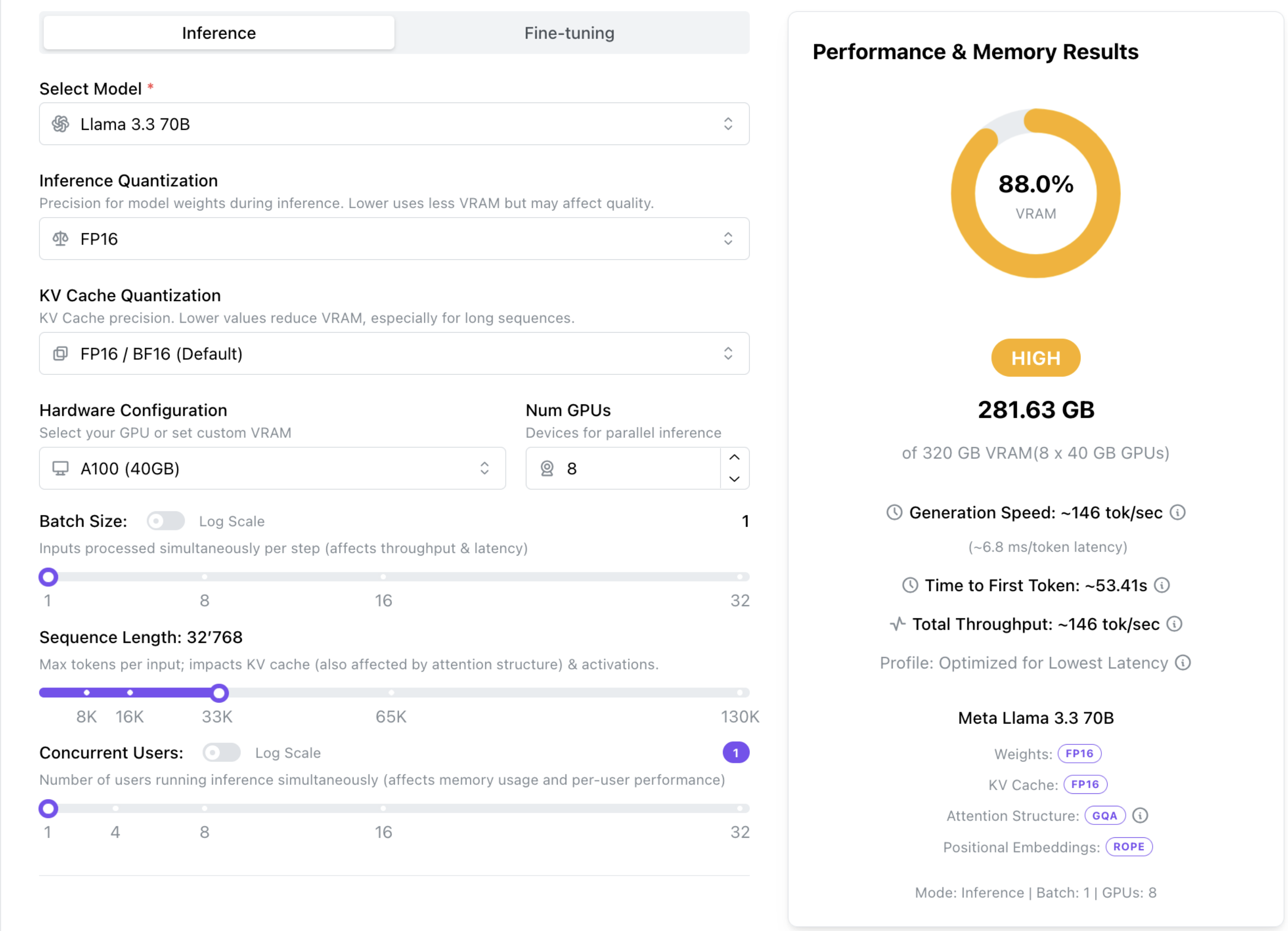
- Llama 3.3 70B is only available globally
- Calculation with PAYG offering
- Quantization is not known

Models	Input (Per 1,000 tokens)	Output (Per 1,000 tokens)
Llama 3.3 70B Datazone	N/A	N/A
Llama 3.3 70B Global	CHF 0.000567	CHF 0.000567
Llama 3.3 70B Regional	N/A	N/A
Llama4 Maverick 17B Datazone	N/A	N/A
Llama4 Mavrick 17B Global	CHF 0.000200	CHF 0.000799
Llama4 Maverick 17B Regional	N/A	N/A

https://azure.microsoft.com/en-us/pricing/details/ai-foundry-models/llama/

Example: Calculate Cost for LLM Hosting – Azure VMs

– FP16 is quite high, lower quantization would increase performance but reduce accuracy



https://apxml.com/tools/vram-calculator

Example: Calculate Cost for LLM Hosting – Azure VMs

Instance	vCPU(s)	RAM	Temporary storage	GPU	Pay as you go	1 year savings plan	3 year savings plan	Spot	Add to estimate
ND96asr A100 v4	96	900 GiB	6,500 GiB	8x A100 (NVlink)	CHF 20,611.0626/ month	CHF 17,142.2226/ month ~16% savings	CHF 10,326.1441/ month ~49% savings	CHF 4,534.4338/ month ~78% savings	+

- ND96asr VM SKU with 8x A100 (40GB), 320GB GPU memory in total (EU west)
- Llama 3.3 70B weights loaded via Huggingface and inference through vllm
- Not included: Network, disk and other operational costs
- Context window 32k (reduce memory usage), FP16 Quantization
- About 146 tokens per second, with Q8 ~191 tokens per second possible

https://azure.microsoft.com/en-us/pricing/details/virtual-machines/linux/#pricing

Example: Calculate Cost for LLM Hosting – Summary

Scenario	Description	VM 3 Year Saving Plan	Inference Endpoint (global)
Maximum token output per hour	Theoretical limit	146 tokens/sec = 525’600 tokens per hour. CHF 14.14 per hour	Virtually unlimited tokens. No cost per hour if no tokens generated (PAYG)
Cost per 1k tokens	Aligned cost per token	CHF 0.0269025875 per 1k tokens	0.000567 CHF per 1k input tokens 0.000567 CHF per 1k output tokens Blended price 0.0008505 per 1k tokens
Cost per month scenario	255'500’000 tokens required	CHF 6'873.61, but actually CHF 10’326.14	CHF 217.30

1 hour = 3600 seconds, 1 month = 730 hours, assuming 2:1 ratio (1k input tokens, 500 output tokens)

Why are LLMs so resource intense?

- Model Weights (Billions), i.e. 70B * FP16 (2 byte per parameter) = 140 GB of VRAM required only for inference
- Each input prompt requires extensive matrix calculations
  - Prefill – load prompt tokens into VRAM and calculate KV vectors
  - Decode – Generate sequentially one token at a time, leveraging the KV cache.

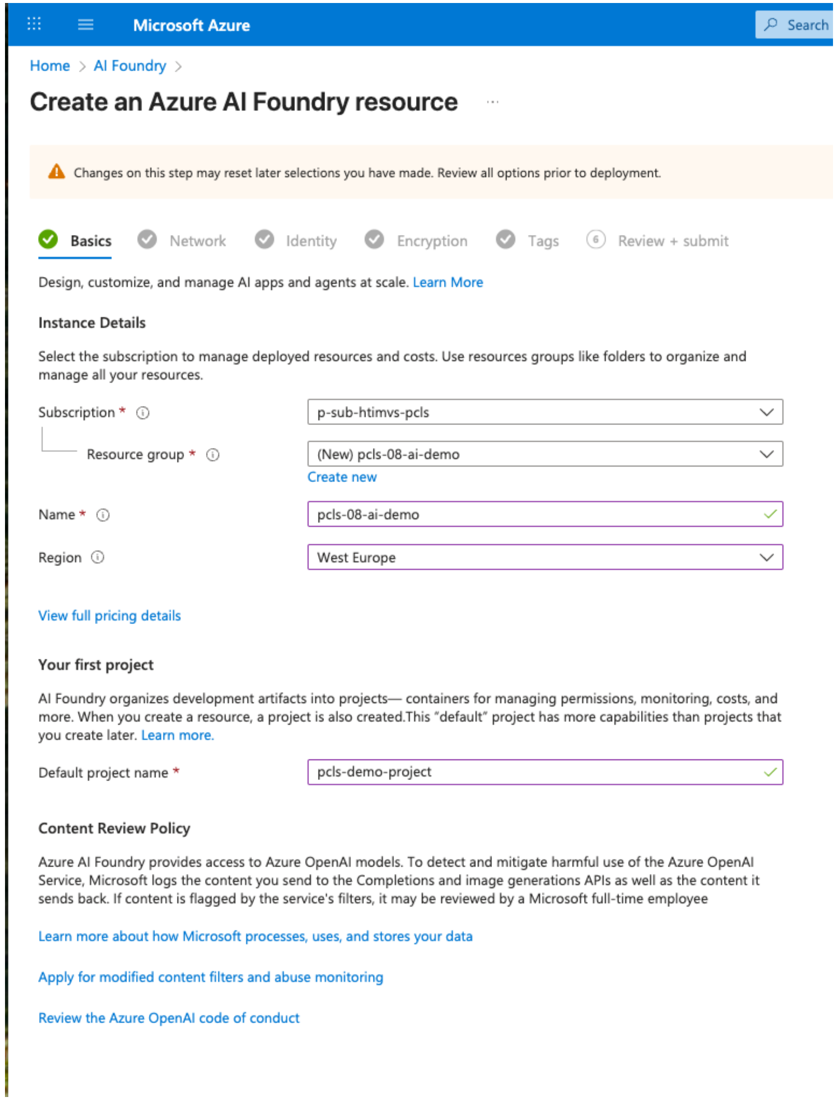
Model Name	Training Time (GPU hours)	Training Power Consumption (W)	Training Location-Based Greenhouse Gas Emissions (tons CO2eq)	Training Market-Based Greenhouse Gas Emissions (tons CO2eq)
Llama 4 Scout	5.0M	700	1,354	0
Llama 4 Maverick	2.38M	700	645	0
Total	7.38M	-	1,999	0

https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E

Demo

Creating an OpenAI LLM endpoint

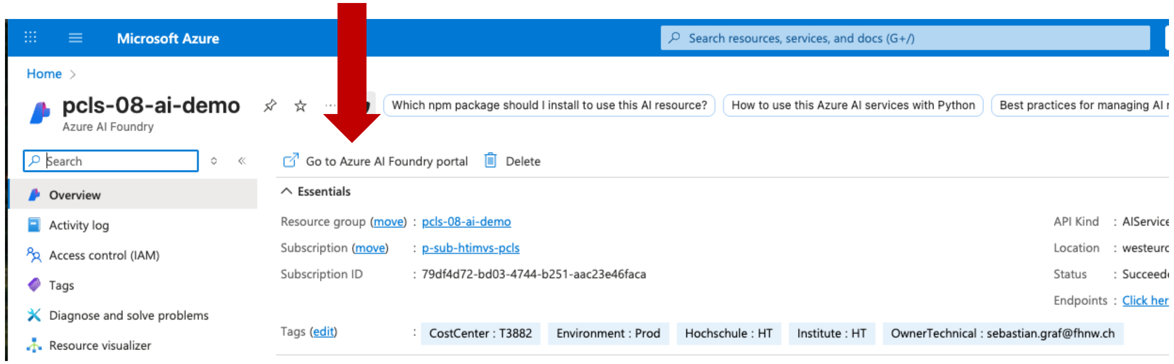
- Log in to https://portal.azure.com
- Navigate to your own resource group
- Create a new resource – search for Azure AI Foundry
- Create a new AI Foundry instance

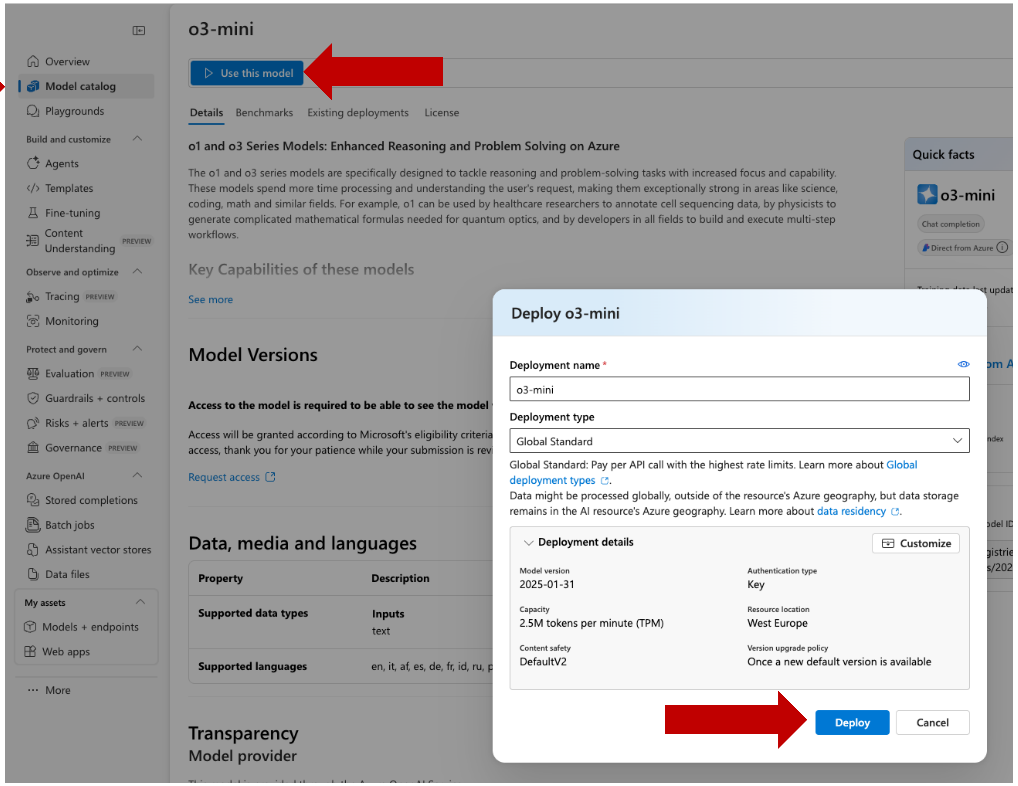


Demo

Provision a model

- Navigate to the AI Foundry instance
- Model Catalog – Search for “o3-mini”
- Deploy model (Deployment type “Global Standard”)

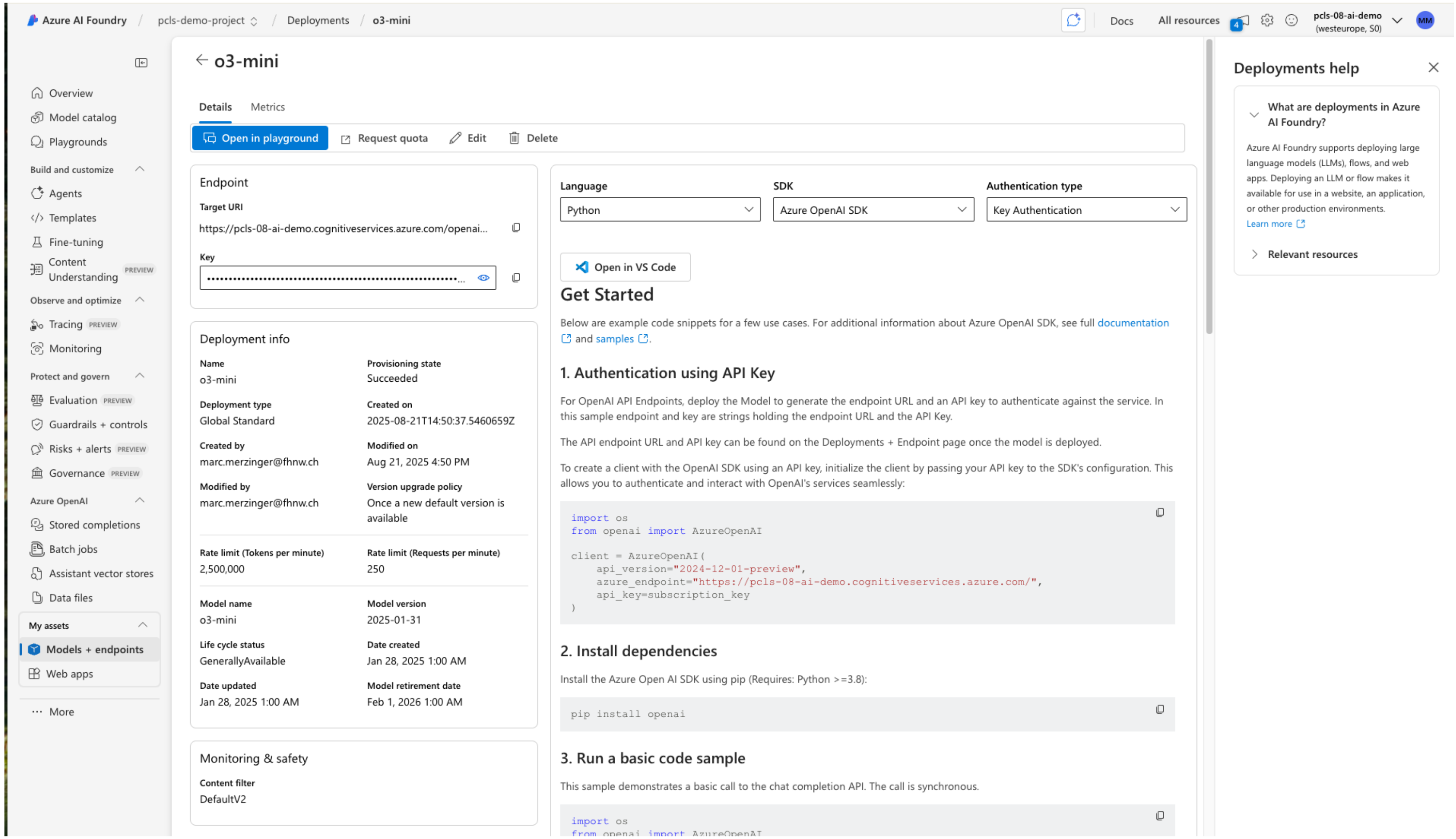




Demo

Final Model-Deployment

- Tests via “Open in playground”
- Integration into code, see examples



Demo WrapUp

Features of the OpenAI LLM endpoint on Azure

- Own logical endpoint
- Control over network location (e.g., in VNET) and authorization/authentication
- Shared GPU and model inference
- Inferences from multiple Azure customers run on the same GPUs (logical isolation)
- Inference takes place globally, i.e., usually where there is the least load on the servers
- Payment based on input and output tokens (regardless of how “long” the service runs)

Alternative Inference Endpoint Deployments

Dedicated Endpoints

- Dedicated model deployment in selected region
- GPUs are dedicated and not shared with other endpoints
- Full control options, but still fully managed by the provider

Self-managed

- Own hardware (purchased or leased)
- Own tech stack (e.g., Ubuntu, Docker, vllm)
- Full control and overall responsibility

Inference API Specifications

Conversation APIs for LLMs

- De facto standard “chat/completions” from OpenAI
- Some manufacturers offer their own endpoints, e.g., Anthropic with “v1/complete”
- Almost all providers support OpenA-compatible endpoints

v1/completions
Legacy Endpoint
Text-based

v1/chat/completions
De facto standard
Supports text, images, or files as input

v1/embeddings
Calculation of embedding vectors of texts

v1/generations
Generation of images (text-to-image)

Inference APIs

```
curl --request POST --url 'https://YOUR-FOUNDRY-RESOURCE-NAME.services.ai.azure.com/openai/deployments/gpt-4o/chat/completions
-h 'authorization: Bearer $AZURE_AI_AUTH_TOKEN' \
-h 'content-type: application/json' \
-d '{
        "messages": [
            {"role": "system",
            "content": "You are a helpful writing assistant"},
            {"role": "user",
            "content": "Write me a poem about flowers"}
        ],
        "model": "gpt-4o"
    }'
```

- Completions/inference requests are sent using a POST request.
- Each message is labeled with a role (system, assistant, user).
- Messages can also contain content parts (e.g., text and files).

https://learn.microsoft.com/en-us/azure/ai-foundry/quickstarts/get-started-code?tabs=rest&pivots=fdp-project

AI Platforms and Inference APIs

AI-Platforms

- AWS-Bedrock
- GCP Vertex AI
- Huggingface
- Swisscom AI Platform

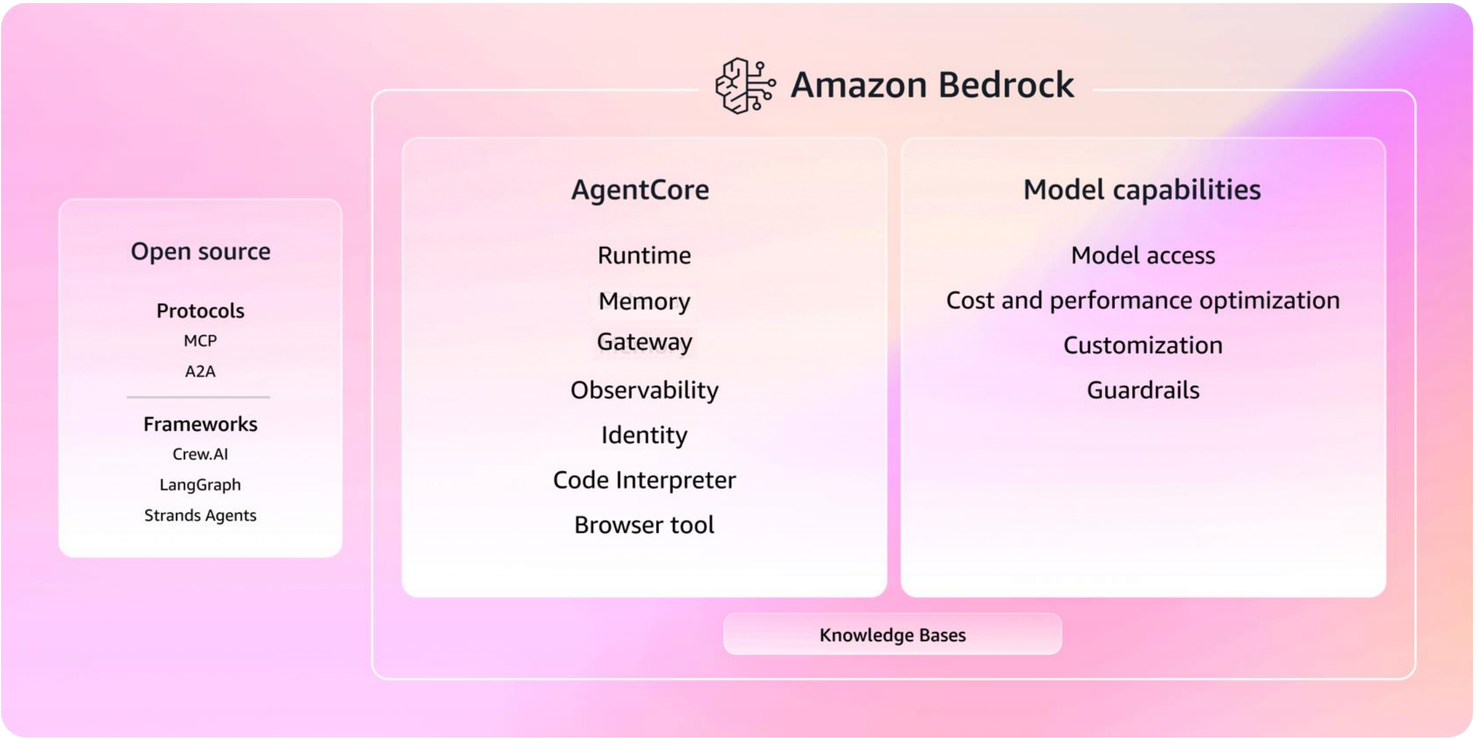
Pure Inference API providers

- OpenRouter
- Together AI

AI – Artificial Intelligence – AWS Bedrock

AI platform from AWS

- Similar to Azure AI Foundry
- Only OpenAI OSS models, e.g., no o4-mini or similar
- Native integration into AWS services such as IAM



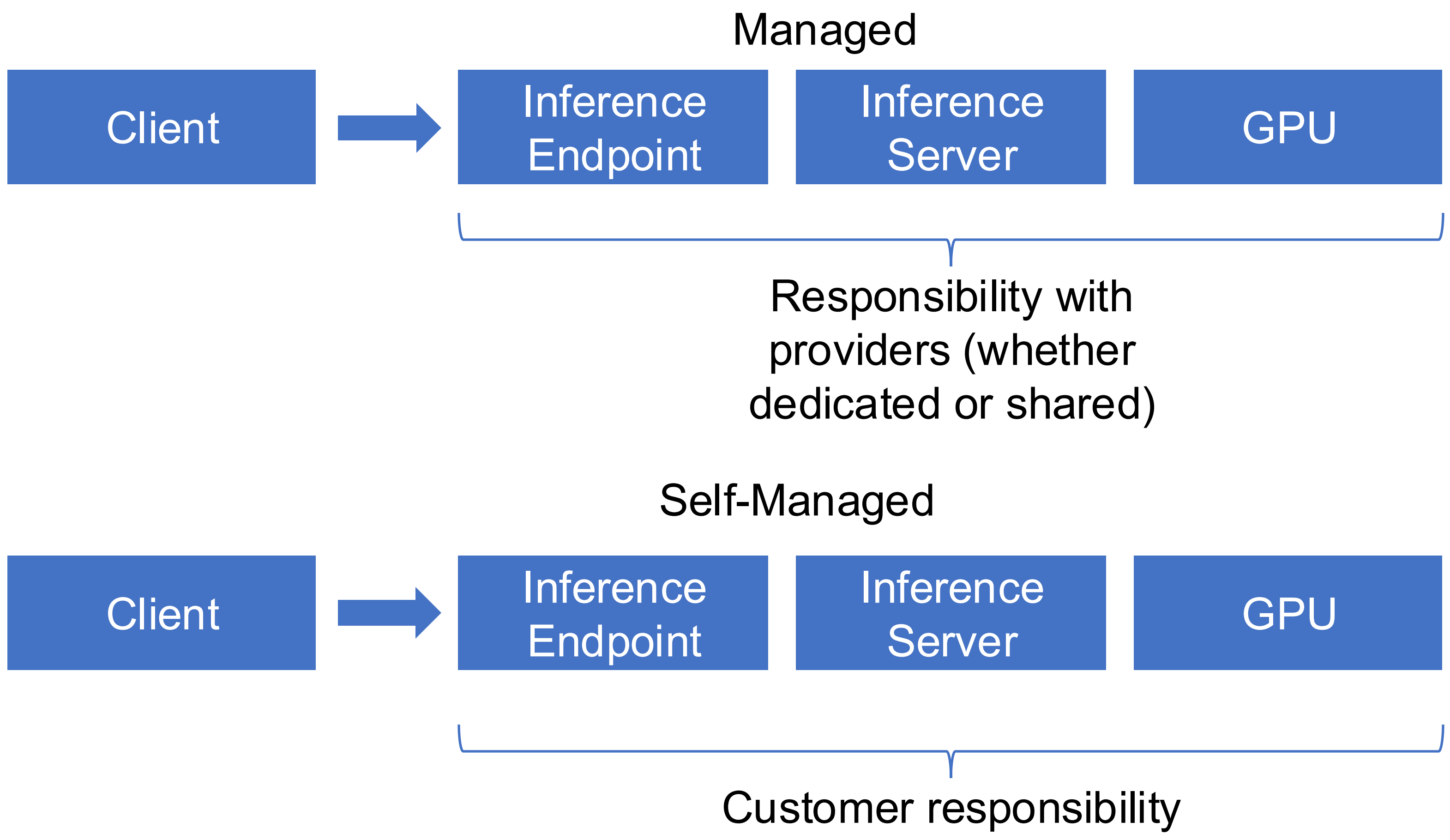
Data Protection and Encryption

Managed LLM Inference

- Encryption of requests in transit via TLS
- Encryption up to GPU optional and dependent on service

Self-Managed LLM

- End-to-end encryption possible up to GPU
- Confidential compute possible



Data Residience

Managed	**Self-Managed**
Inference: - Choice between Global, Data Zone, and Region Limits: - Model availability - Throughput limits - Usually shared infrastructure	Inference: - Self-managed deployments in regions or your own data center Limits: - Self-managed deployments in regions or your own data center

Data Retention and Training Usage – Example Open AI

Open AI vs New York Times

Update on October 22, 2025:

After months of litigation, we are no longer under a legal order to retain consumer ChatGPT and API content indefinitely. Our obligations under the earlier order ended on September 26, 2025.

We've returned to our standard data retention practices:

- ***Deleted ChatGPT conversations** and **Temporary Chats** will be automatically deleted from our systems within 30 days.*
- ***API data** will also be automatically deleted after 30 days.*

https://openai.com/index/response-to-nyt-data-demands/

Open AI User Data Training

Services for individuals, such as ChatGPT, Codex, and Sora

When you use our services for individuals such as ChatGPT, Codex, and Sora, we may use your content to train our models.

You can opt out of training through our privacy portal by clicking on "do not train on my content." To turn off training for your ChatGPT conversations and Codex tasks, follow the instructions in our Data Controls FAQ. Once you opt out, new conversations will not be used to train our models.

Explicit Opt-Out needed!

https://help.openai.com/en/articles/5722486-how-your-data-is-used-to-improve-model-performance

Data Protection – Guardrails and Content-Filter

Guardrails

- Threshold values for offensive language or topics
  - Avoiding inappropriate responses
  - Avoiding aiding and abetting unacceptable behavior
- Resistance to prompt injection or similar
  - Protection of system instructions

Categories

- Hate und Fairness
- Sexual
- Violence
- Self-Harm

https://learn.microsoft.com/en-us/python/api/overview/azure/ai-contentsafety-readme?view=azure-python

Data Protection – Guardrails and Content-Filter

```python
import os
from azure.ai.contentsafety import ContentSafetyClient
from azure.ai.contentsafety.models import TextCategory
from azure.core.credentials import AzureKeyCredential
from azure.core.exceptions import HttpResponseError
from azure.ai.contentsafety.models import AnalyzeTextOptions

key = os.environ["CONTENT_SAFETY_KEY"]
endpoint = os.environ["CONTENT_SAFETY_ENDPOINT"]

# Create a Content Safety client
client = ContentSafetyClient(endpoint, AzureKeyCredential(key))

# Construct a request
request = AnalyzeTextOptions(text="You are an idiot")

# Analyze text
try:
    response = client.analyze_text(request)
except HttpResponseError as e:
    print("Analyze text failed.")
    if e.error:
        print(f"Error code: {e.error.code}")
        print(f"Error message: {e.error.message}")
        raise
    print(e)
    raise
```

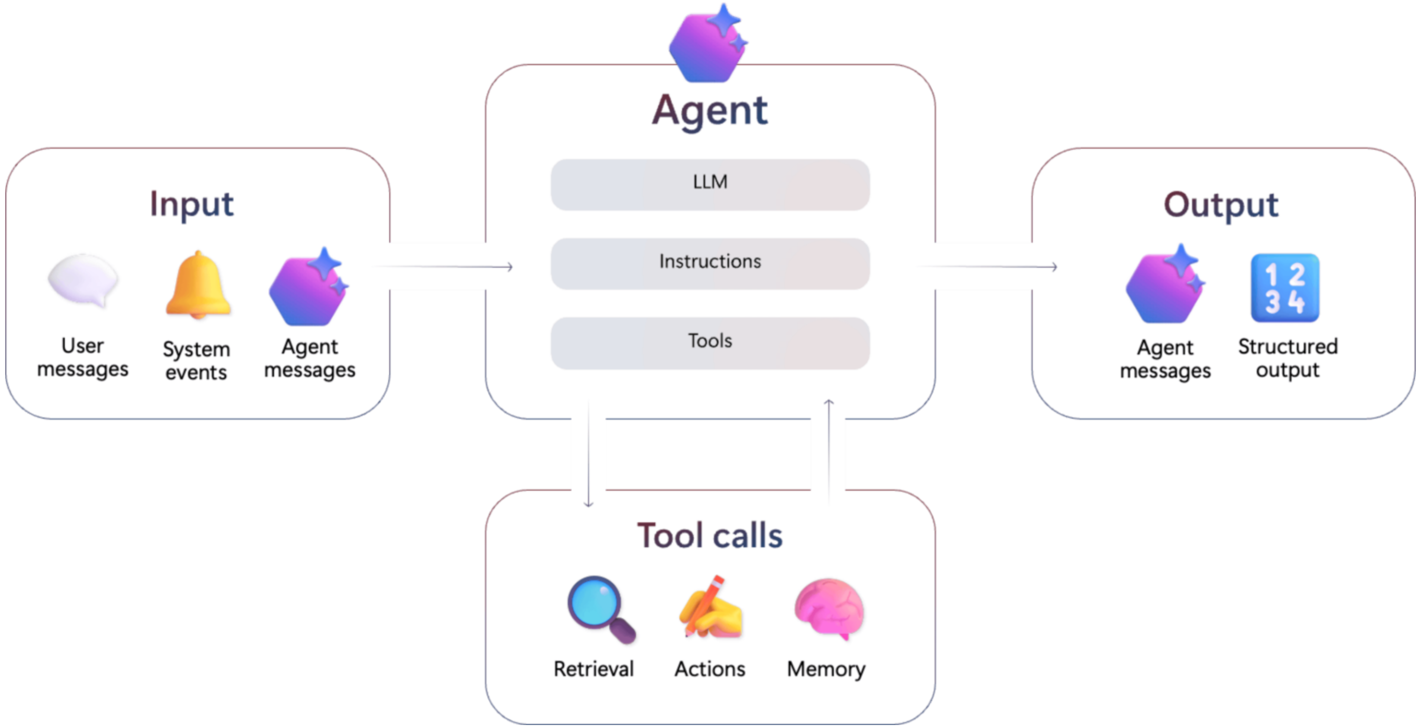
```python
hate_result = next(item for item in response.categories_analysis if item.cate
self_harm_result = next(item for item in response.categories_analysis if item
sexual_result = next(item for item in response.categories_analysis if item.ca
violence_result = next(item for item in response.categories_analysis if item.

if hate_result:
    print(f"Hate severity: {hate_result.severity}")
if self_harm_result:
    print(f"SelfHarm severity: {self_harm_result.severity}")
if sexual_result:
    print(f"Sexual severity: {sexual_result.severity}")
if violence_result:
    print(f"Violence severity: {violence_result.severity}")
```

https://learn.microsoft.com/en-us/python/api/overview/azure/ai-contentsafety-readme?view=azure-python#analyze-text-without-blocklists

AI-Agents

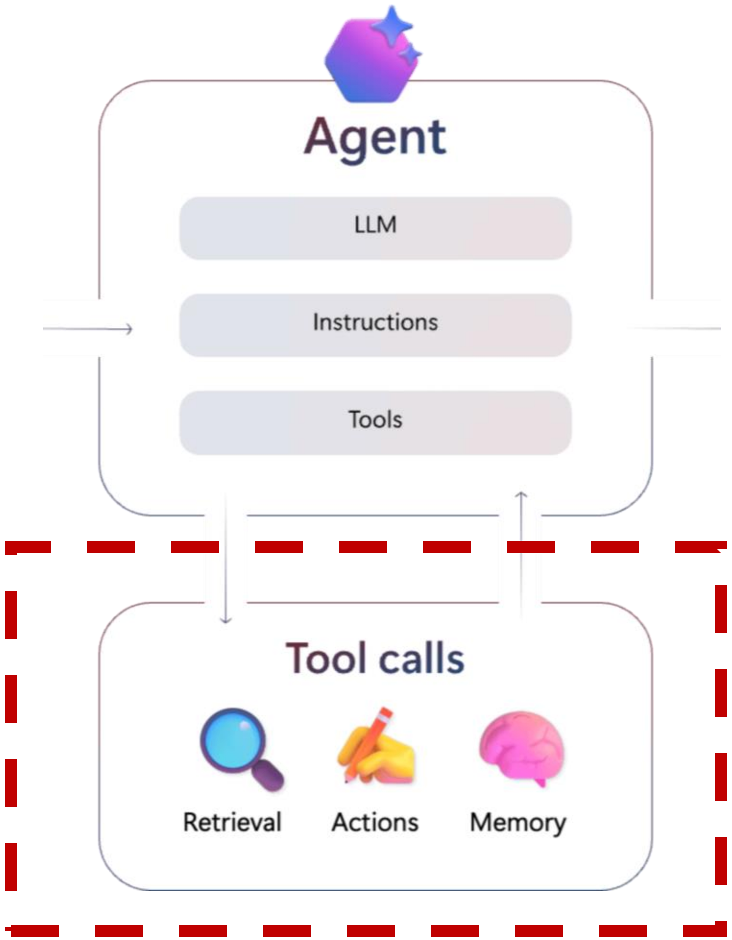
- AI agents are wrappers around LLMs
- More context and integration possible through APIs
- Model Context Protocol (MCP)
Protocol for standardizing LLM to tool communication
- Agent to Agent (A2A) Protocol in development



https://learn.microsoft.com/en-us/azure/ai-foundry/agents/overview

Agents Toolcalling

- Tools include Rest APIs or MCP servers, for example.
- LLM recognizes the need for a tool based on input and outputs the response using a special format.
- The agent executes the tool and sends a new prompt (with tool results and original user prompt) to the LLM. The answer is then returned to the user.



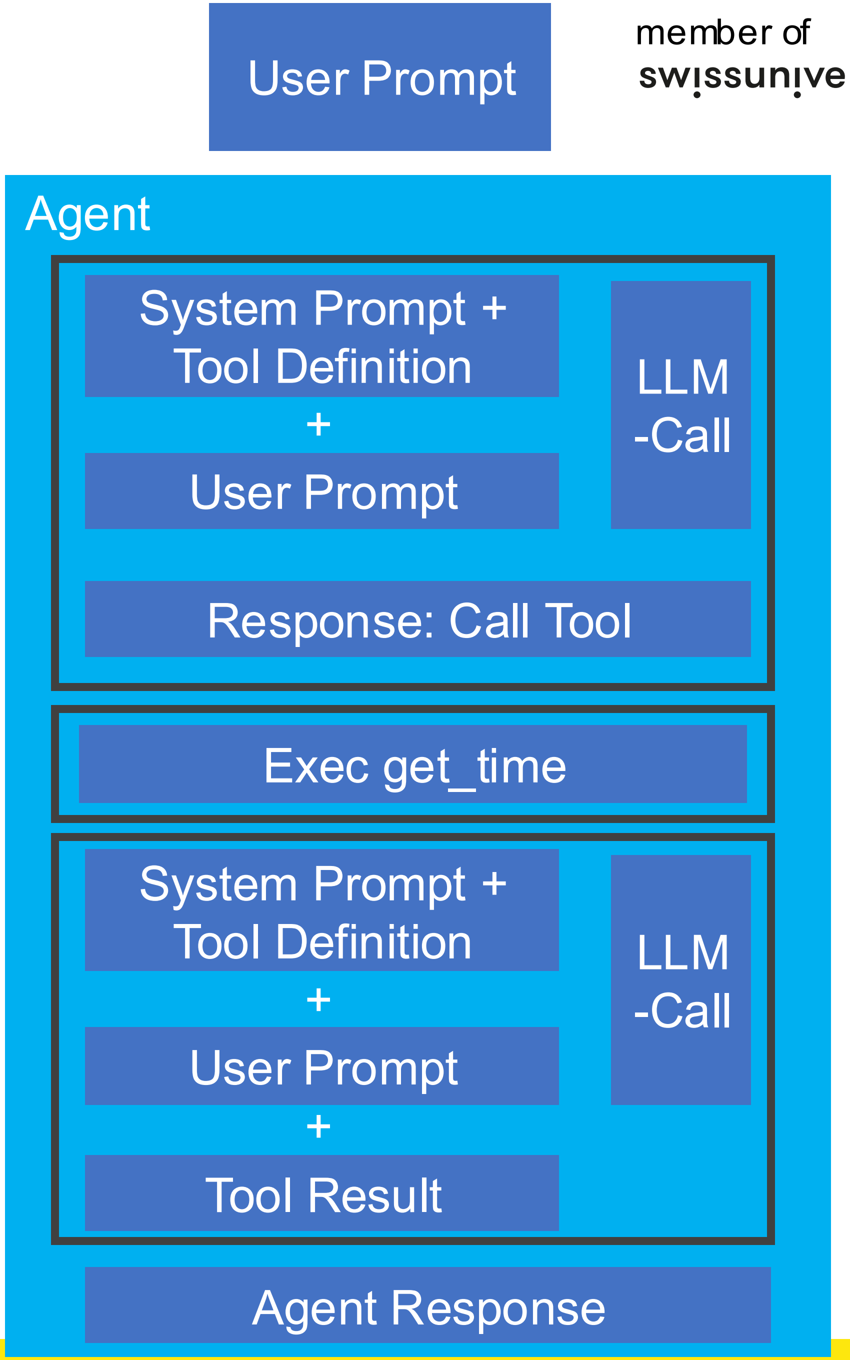
Agent Toolcalling

System Prompt:

You are a helpful assistant who can always tell the user the current time. No matter what the user wants, always tell them the current time. Use the get_time tool to find out the time.

Available tools:

get_time: Retrieves the current time.



Agent Frameworks

- Agent frameworks offer
  - Standardized interfaces for interacting with LLMs (interchangeability)
  - Wrappers and structures for (automated) tool calls
  - Integrations with other standards such as OpenTelemetry
  - Standard patterns such as workflow agents
- Python is the most popular programming language for agents

Autogen

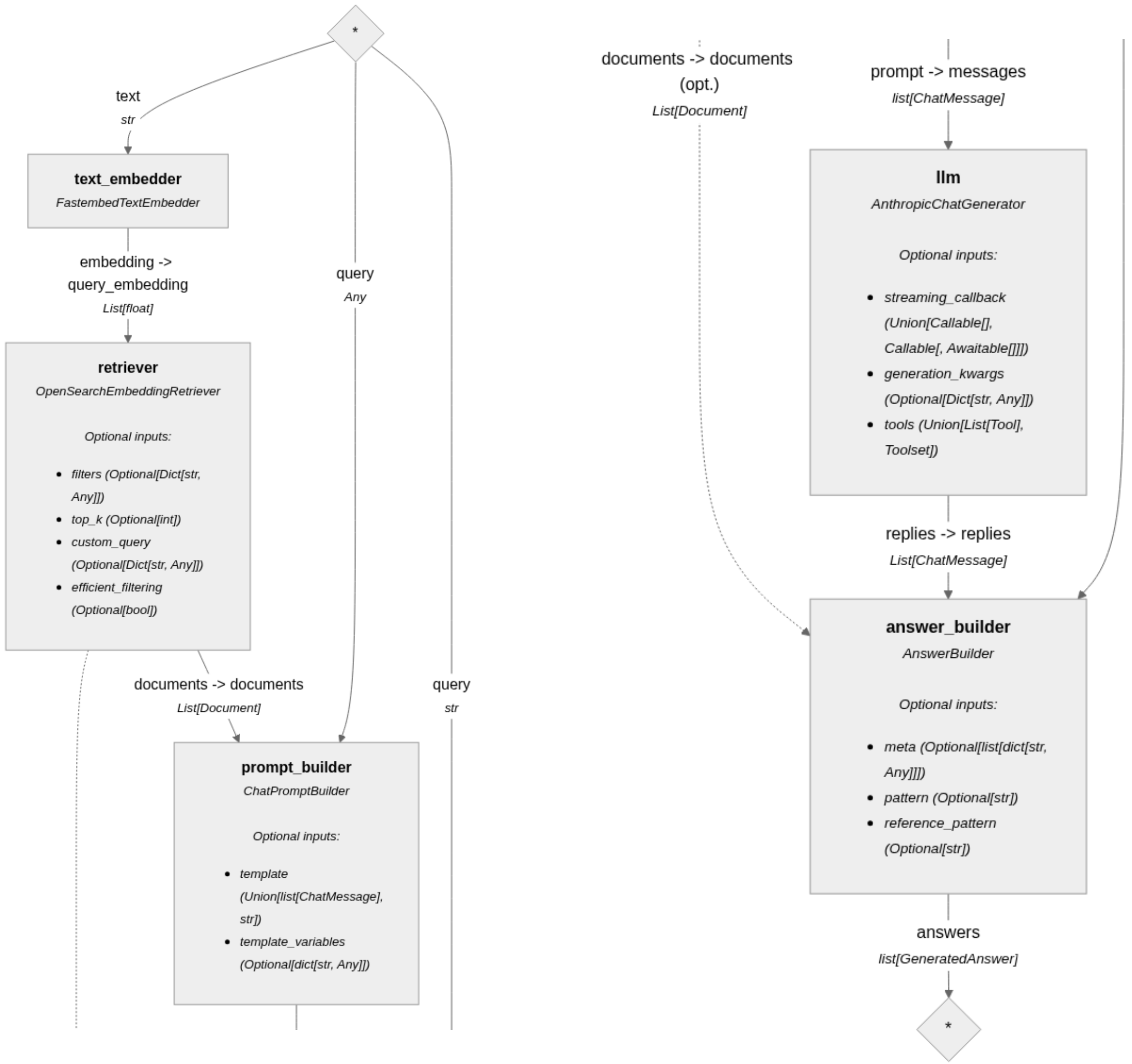
crewai

Langchain

Google ADK

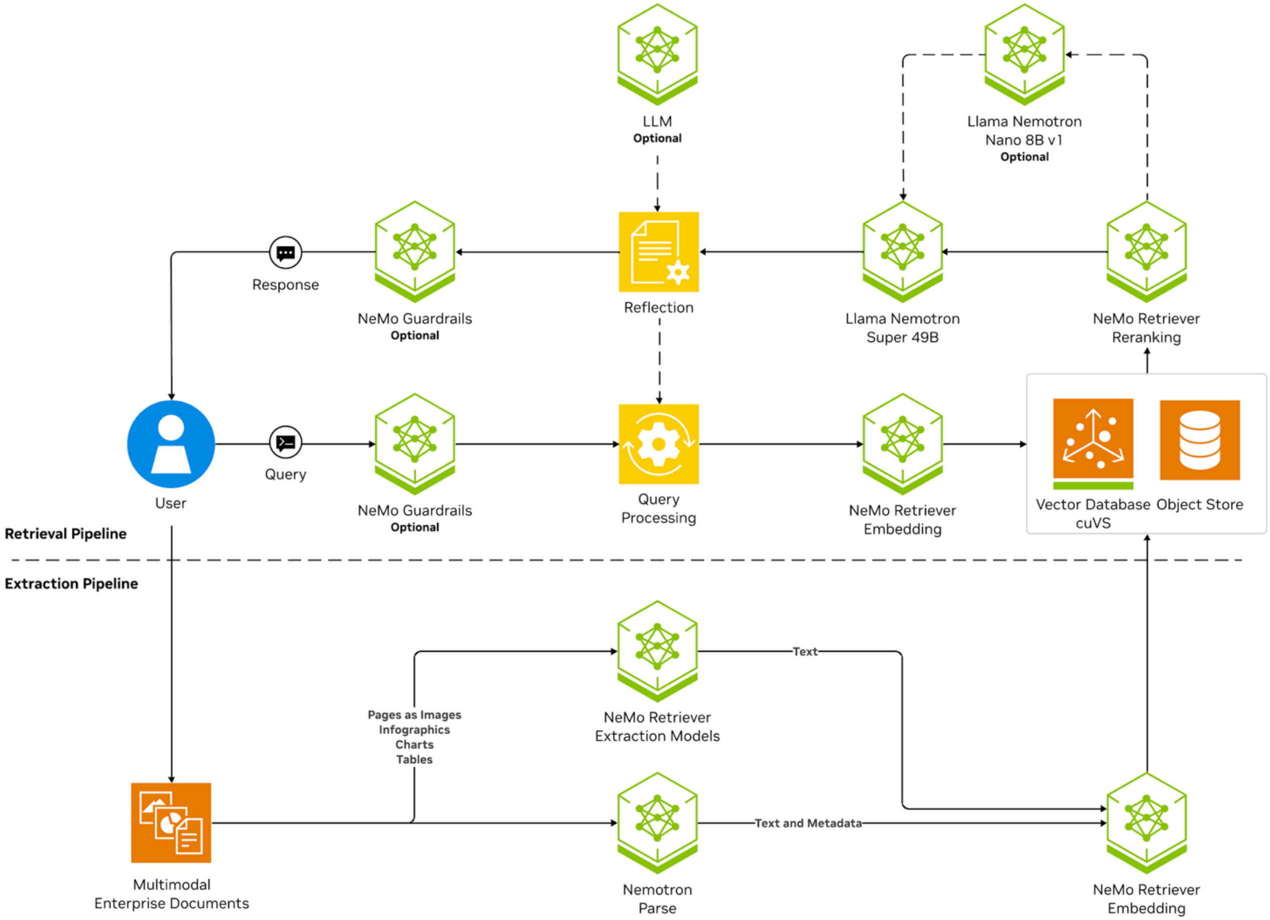
AI Search / RAG

- Search service for your own data
- Indexing and sharding are performed via the service and exposed as an API for use by your own application
- Option to restrict permissions on documents
- RAG is one of several supported use cases of AI Search



https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search

RAG Reference Architecture



https://www.nvidia.com/en-us/glossary/retrieval-augmented-generation/

Why RAG is needed

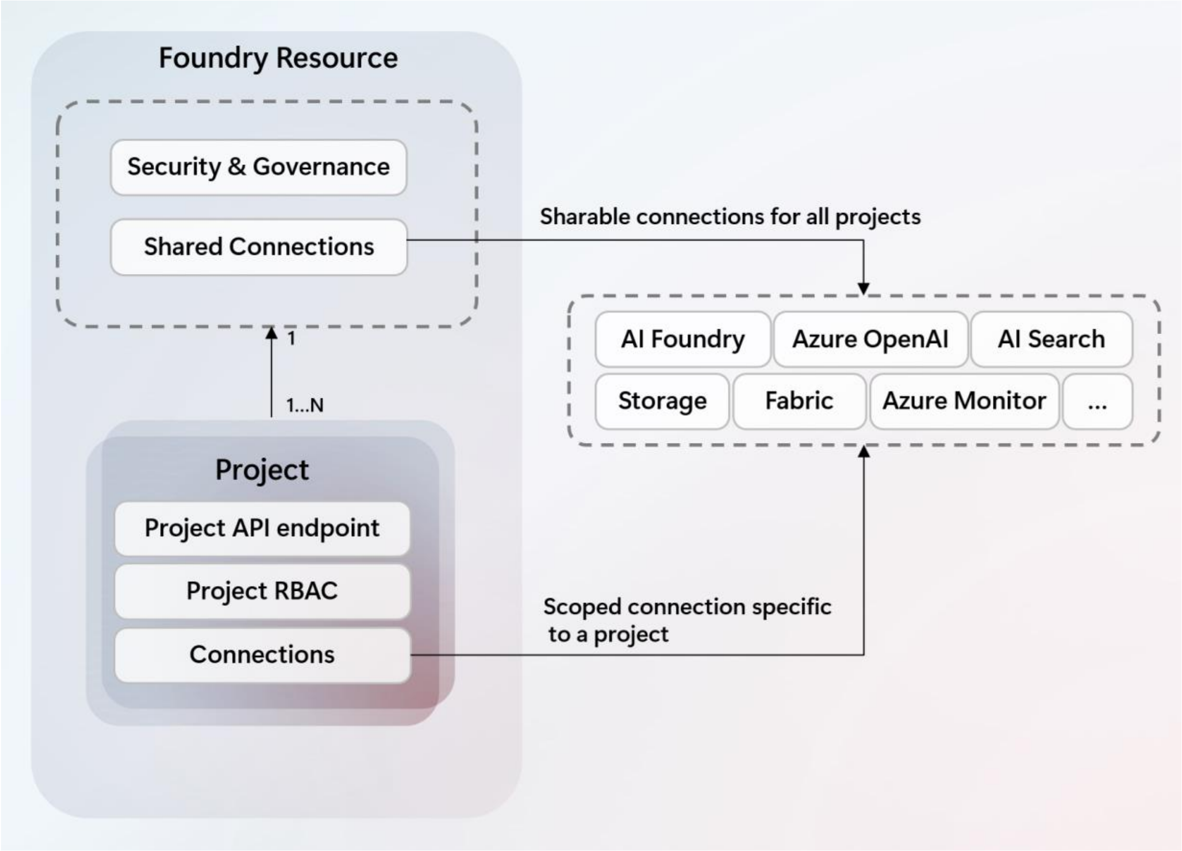
- LLMs memorized their training data, but are not aware of company internal data
- Prompts are limited by the Context Window size (i.e. 128k tokens) – which fits only a few documents
- The longer the context the lower the accuracy gets
- Fine-tuning is resource and know-how intensive
- Fine-tuning does not allow the enforcement of RBAC or other access restrictions

Model Name	Acc avg	4k	8k	16k	32k	64k	96k	128k	19
Qwen3-30B-A3B-Instruct-2507	86.8	98.0	96.7	96.9	97.2	93.4	91.0	89.1	89
Qwen3-235B-A22B-Instruct-2507	92.5	98.5	97.6	96.9	97.3	95.8	94.9	93.9	94
Qwen3-Next-80B-A3B-Instruct	91.8	98.5	99.0	98.0	98.7	97.6	95.0	96.0	94

https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct
Qwen3 accuracy reduction with larger context sizes

Infrastructure as Code for Azure AI Foundry

- The resources of the AI Foundry are provided by the resource provider Microsoft.CognitiveServices
- Central structural element “project” for endpoints, permissions, etc., so that multiple people can work on the same resource



https://learn.microsoft.com/en-us/azure/ai-foundry/concepts/planning

Infrastructure as Code for Azure AI Foundry

- Bicep example of an Azure AI Foundry resource (aka Microsoft.CognitiveServices/account)
- SKU: Determines which types are supported in this account. For example, S0 supports AIServices, ContentSafety, or FormRecognizer (and others).
- Kind: One of the types supported by the SKU, such as AIServices.

```
resource aiFoundry 'Microsoft.CognitiveServices/accounts@2025-04-01-preview' = {
  name: aiFoundryName
  location: location
  identity: {
    type: 'SystemAssigned'
  }
  sku: {
    name: 'S0'
  }
  kind: 'AIServices'
  properties: {
    // required to work in AI Foundry
    allowProjectManagement: true

    // Defines developer API endpoint subdomain
    customSubDomainName: aiFoundryName

    disableLocalAuth: true
  }
}
```

https://github.com/azure-ai-foundry/foundry-samples/blob/main/samples/microsoft/infrastructure-setup/00-basic/main.bicep

Infrastructure as Code for Azure AI Foundry

- Bicep example of a model deployment (aka Microsoft.CognitiveServices/accounts/deployment)
- Sku: Determines the capacity and location of the model deployment
- Properties: Specifies the model

```
resource modelDeployment 'Microsoft.CognitiveServices/accounts/deployments@2024-10-01'= {
  parent: aiFoundry
  name: 'gpt-4o'
  sku : {
    capacity: 1
    name: 'GlobalStandard'
  }
  properties: {
    model:{
      name: 'gpt-4o'
      format: 'OpenAI'
    }
  }
}
```

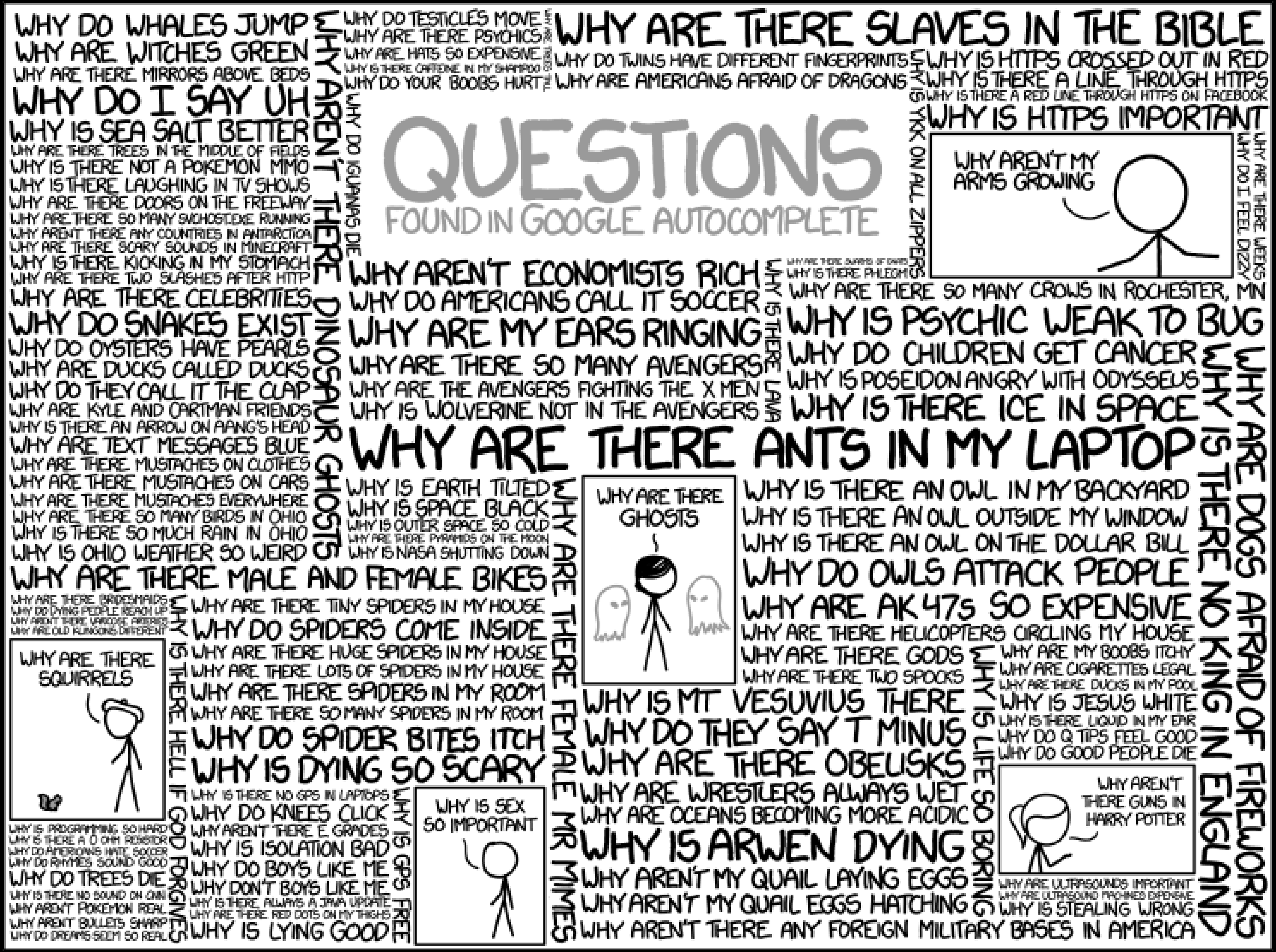
https://github.com/azure-ai-foundry/foundry-samples/blob/main/samples/microsoft/infrastructure-setup/00-basic/main.bicep

Summary

- Cloud providers' AI services cover the entire lifecycle from ML to AI (agent) use cases
- Large commercial large language models have limited availability in the clouds (regions, model selection, capacities)
- LLM training and inference are resource-intensive
- AI agents use LLMs as a decision-making component to execute actions (e.g., Google search or document upload)

Homework

- Review the material from this lesson.
- Complete Assignment 08.
- Create and deploy your own AI service instances with the following learning paths (if needed):
  - Introduction to AI in Azure (Learning Path) (quite a lot of content)
  - Generative AI Modul


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY AREN'T THERE DINOSAUR GHOSTS
WHY DO IGUANAS DIE
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS YKK ON ALL ZIPPERS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS THERE LAVA
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY IS THERE NO KING IN ENGLAND
WHY ARE DOGS AFRAID OF FIREWORKS
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE ARTERIES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS THERE HELL IF GOD FORGIVES
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY ARE THERE FEMALE MR MIMES
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS LIFE SO BORING
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY IS GPS FREE
WHY IS SEX SO IMPORTANT